Exercícios Programados 7 - Gabarito

1^a **Questão:** Mostre que $exp(3 + 7\pi i) = -e^3$.

Solução:

$$exp(3 + 7\pi i) = e^3(cos(7\pi) + isen(7\pi)) = e^3(-1 + i0) = -e^3$$

2^a **Questão:** Escreva os complexos abaixo na forma polar $re^{i\theta}$.

(a) $z = 1 + \sqrt{3}i$ **(c)** $z = -\sqrt{3} + i$

(b) $z = -2 + 2i$ **(d)** $z = -\frac{1}{8} - \frac{1}{8}i$

Solução:

(a)

$$r = \sqrt{1^2 + \left(\sqrt{3}\right)^2} = \sqrt{4} = 2$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z)}{r} = \dfrac{1}{2} \\ \sin(\theta) = \dfrac{Im(z)}{r} = \dfrac{\sqrt{3}}{2} \end{cases} \implies \theta = \frac{\pi}{3} \implies z = 2e^{\frac{\pi}{3}i}$$

(b)

$$r = \sqrt{(-2)^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z)}{r} = \dfrac{-2}{2\sqrt{2}} = \dfrac{-\sqrt{2}}{2} \\ \sin(\theta) = \dfrac{Im(z)}{r} = \dfrac{2}{2\sqrt{2}} = \dfrac{\sqrt{2}}{2} \end{cases} \implies \theta = \frac{3\pi}{4} \implies z = 2\sqrt{2}e^{\frac{3\pi}{4}i}$$

(c)

$$r = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = \sqrt{4} = 2$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z)}{r} = \dfrac{-\sqrt{3}}{2} \\ \sin(\theta) = \dfrac{Im(z)}{r} = \dfrac{1}{2} \end{cases} \implies \theta = \frac{5\pi}{6} \implies z = 2e^{\frac{5\pi}{6}i}$$

(d)

$$r = \sqrt{\left(-\frac{1}{8}\right)^2 + \left(-\frac{1}{8}\right)^2} = \sqrt{\frac{1}{32}} = \frac{1}{4\sqrt{2}}$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z)}{r} = \dfrac{-\sqrt{2}}{2} \\ \sin(\theta) = \dfrac{Im(z)}{r} = \dfrac{-\sqrt{2}}{2} \end{cases} \implies \theta = \frac{5\pi}{4} \qquad \implies z = \frac{1}{4\sqrt{2}} e^{\frac{5\pi}{4}i}$$

3ª **Questão:** Prove as fórmulas de Euler:

$$cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \qquad sen\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Solução:

$$\begin{aligned} \frac{e^{i\theta} + e^{-i\theta}}{2} &= \frac{((cos\theta + isen\theta) + ((cos(-\theta) + isen(-\theta)))}{2} \\ &= \frac{((cos\theta + isen\theta) + ((cos(\theta) - isen(\theta)))}{2} = \frac{2cos\theta}{2} = cos\theta \\ \frac{e^{i\theta} - e^{-i\theta}}{2i} &= \frac{((cos\theta + isen\theta) - ((cos(-\theta) + isen(-\theta)))}{2i} \\ &= \frac{((cos\theta + isen\theta) - ((cos(\theta) - isen(\theta)))}{2i} = \frac{2isen\theta}{2i} = sen\theta \end{aligned}$$

4ª **Questão:** Prove que

$$1 + cos(\theta) + cos(2\theta) + ... + cos(n\theta) = \frac{1}{2} + \frac{sen[(n + 1/2)]\theta}{2sen(\theta/2)}$$

e

$$sen(\theta) + sen(2\theta) + ... + sen(n\theta) = \frac{cos(\theta/2) - cos[(n + 1/2)]\theta}{2sen(\theta/2)}$$

Sugestão: Desenvolva $\sum_{k=0}^{n}(\cos(k\theta) + i\sin(k\theta)) = \sum_{k=0}^{n} e^{ik\theta}$ usando a soma de P.G. e a Fórmula de De Moivre

Solução: Pela soma de P.G. de razão $e^{i\theta}$, tem-se que:

$$\sum_{k=0}^{n}(\cos(k\theta) + i\sin(k\theta)) = \sum_{k=0}^{n} e^{ik\theta} = \frac{1 - e^{i(n+1)\theta}}{1 - e^{i\theta}}$$

Mulitplicando o numerador e o denominador por $e^{-i\theta/2}$, segue que:

$$\frac{1-e^{i(n+1)\theta}}{1-e^{i\theta}} = \frac{e^{-i\theta/2}-e^{i(n+1/2)\theta}}{e^{-i\theta/2}-e^{i\theta/2}}$$

Por outro lado,

$$\begin{aligned}
\frac{e^{-i\theta/2}-e^{i(n+1/2)\theta}}{e^{-i\theta/2}-e^{i\theta/2}} &= \frac{cos(\theta/2)-isen(\theta/2)-cos[(n+1/2)]\theta-isen[(n+1/2)]\theta}{-2isen(\theta/2)} \\
&= \frac{-isen(\theta/2)-isen[(n+1/2)]\theta}{-2isen(\theta/2)} + \frac{cos(\theta/2)-cos[(n+1/2)]\theta}{-2isen(\theta/2)} \\
&= \frac{1}{2} + \frac{sen[(n+1/2)]\theta}{2sen(\theta/2)} + i\left[\frac{cos(\theta/2)-cos[(n+1/2)]\theta}{2sen(\theta/2)}\right]
\end{aligned}$$

Dessa forma, tem-se

$$\sum_{k=0}^{n}(\cos(k\theta)+i\sin(k\theta)) = \frac{1}{2} + \frac{sen[(n+1/2)]\theta}{2sen(\theta/2)} + i\left[\frac{cos(\theta/2)-cos[(n+1/2)]\theta}{2sen(\theta/2)}\right]$$

e, portanto,

$$1+cos(\theta)+cos(2\theta)+...+cos(n\theta) = \frac{1}{2} + \frac{sen[(n+1/2)]\theta}{2sen(\theta/2)}$$

e

$$sen(\theta)+sen(2\theta)+...+sen(n\theta) = \frac{cos(\theta/2)-cos[(n+1/2)]\theta}{2sen(\theta/2)}$$